

*****Default*****

Upgrading openjpeg lib

QEMU RPi RPi2

06/08/26

*****Default*****

```
devel@pi4-50:~ $ ./qemu.sh
```

```
#!/bin/bash
```

```
export QEMU_PATH=$HOME/qemu-6.2.0-rpios/bin
```

```
export PATH=$QEMU_PATH:$PATH
```

```
echo $PATH
```

```
devel@pi4-50:~ $ cd Ultibo_Projects/jpeg2000/src/
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/src $
```

```
modified: dwt.c
```

```
modified: dwt.h
```

```
modified: j2k.c
```

```
modified: j2k.h
```

```
modified: jp2.c
```

```
modified: jp2.h
```

```
modified: mct.c
```

```
modified: mct.h
```

```
modified: mqc.c
```

```
modified: mqc.h
```

```
modified: openjpeg.c
```

```
modified: openjpeg.h
```

```
modified: pi.c
```

```
modified: t1.c
```

```
modified: t1.h
```

```
modified: t2.c
```

```
modified: t2.h
```

```
modified: tcd.c
```

```
modified: tcd.h
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/src $ ./compile_ultibo.sh
```

```
#!/bin/bash
```

```
rm -f *.o
```

```
rm -f libopenjp2.a
```

```
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -  
c bio.c -o bio.o
```

```
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -  
c cio.c -o cio.o
```

```
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -  
c dwt.c -o dwt.o
```

```
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -  
c event.c -o event.o
```

```
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -  
c function_list.c -o function_list.o
```

```
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c image.c -o image.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c invert.c -o invert.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c j2k.c -o j2k.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c jp2.c -o jp2.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c mct.c -o mct.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c mqc.c -o mqc.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c opj_clock.c -o opj_clock.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c opj_malloc.c -o opj_malloc.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c pi.c -o pi.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c t1.c -o t1.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c rd-wr-ops.c -o rd-wr-ops.o
```

```
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c t2.c -o t2.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c tcd.c -o tcd.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c tgt.c -o tgt.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c thread.c -o thread.o
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c openjpeg.c -o openjpeg.o
```

```
arm-none-eabi-gcc -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -
c sparse_array.c -o sparse_array.o
arm-none-eabi-ar rcs libopenjp2.a *.o
arm-none-eabi-ar t libopenjp2.a > libopenjp2_obj.txt
echo "The word count here should be 22"
echo "the word count in /home/pi/jpeg-2000-test/bare-metal/openjp"
echo "when ./libbuild.sh is executed should be 22"
wc libopenjp2_obj.txt
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/RPi2 $ cat libbuild.sh
#!/bin/bash
```

```
rm -f dwtlift.o
rm -f libdwtlift.a
rm -f libopenjp2.a
#cp libopenjp2_816f53.a libopenjp2.a
cp ../src/libopenjp2.a libopenjp2.a
```

```
arm-none-eabi-gcc -L. -llibopenjp2 -O3 -mabi=aapcs -marm -march=armv7-a -mfpu=vfpv3-d16 -mfloat-abi=hard -c dwtlift.c
```

```
arm-none-eabi-ar rcs libopenjp2.a dwtlift.o
```

```
cp libopenjp2.a libdwtlift.a
```

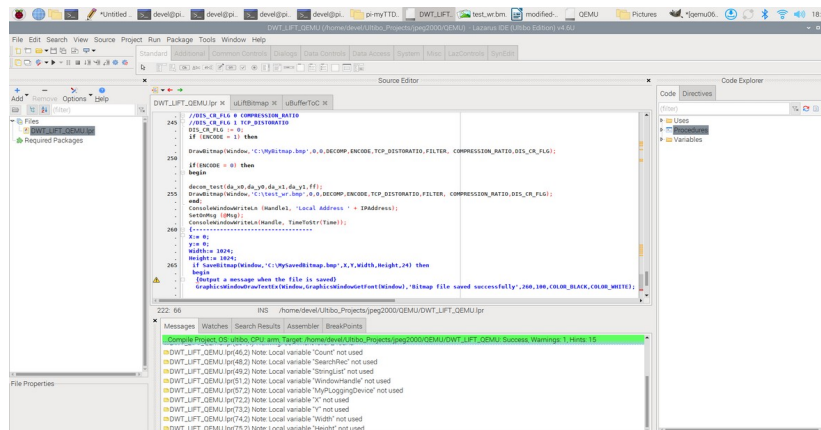
```
arm-none-eabi-ar t libdwtlift.a > libdwtlift_obj.txt
```

```
echo "when ./libbuild.sh is executed should be 23"
```

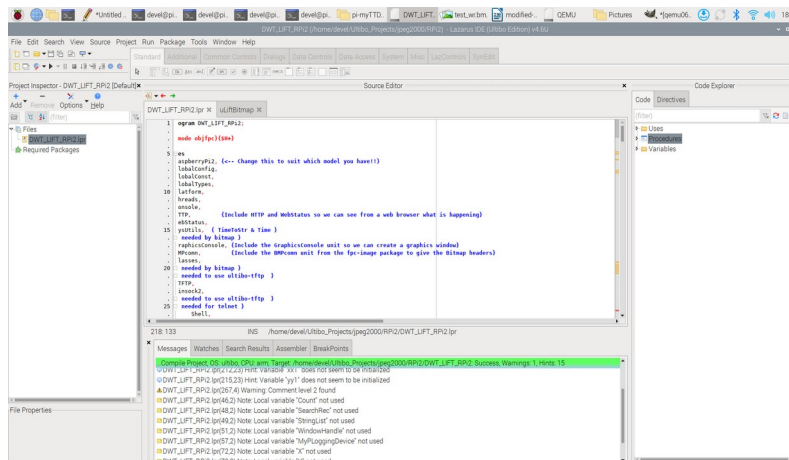
```
wc libdwtlift_obj.txt
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/QEMU $ ./libbuild.sh
```

QEMU

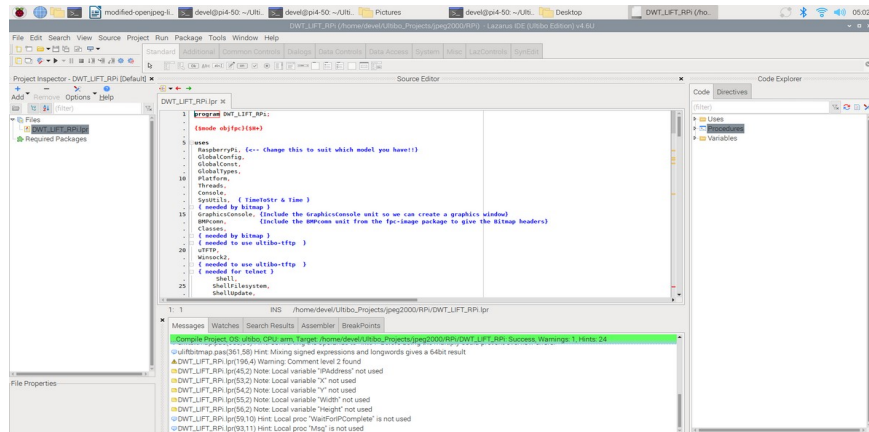


```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/RPi2 $ ./libbuild.sh  
RPi2
```



```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/RPi $ ./libbuild.sh
```

RPi



```
startqemu.sh
#!/bin/bash
qemu-system-arm -machine versatilepb -cpu cortex-a8 -kernel kernel.bin \
-net
user,hostfwd=tcp::5080-:80,hostfwd=tcp::5023-:23,hostfwd=udp::5069-:69,hostfwd=tcp::6050-:505
0 -net nic \
-drive file=disk.img,if=sd,format=raw
```

```
devel@pi4-50:~/Ultibo_Projects/jpeg2000/QEMU $ ./startqemu.sh
```

The image below was following the openjpeg update on 06/08/26

Ultibo Core (Release: Beetroot Version: 2.6.173 Date: 9 February 2026)

```
in output buffer. 35 needed.
[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
in output buffer. 35 needed.
[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
in output buffer. 35 needed.
[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
in output buffer. 35 needed.
[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
in output buffer. 35 needed.
[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
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[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
in output buffer. 35 needed.
[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
in output buffer. 35 needed.
[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
in output buffer. 35 needed.
[ERROR] opj_t2_encode_packet(): only 34 bytes remaining
in output buffer. 35 needed.
Compression time: 474993 seconds 12 useconds 474981
23:15:32
```

```
TFTP Demo.
writing top right handle1
Local Address 10.0.2.15
TFTP Ready.
```



To create the test.j2k the file 256com needs to be copied to testfile. In addition the MyBitmap.bmp is the file to be compressed.

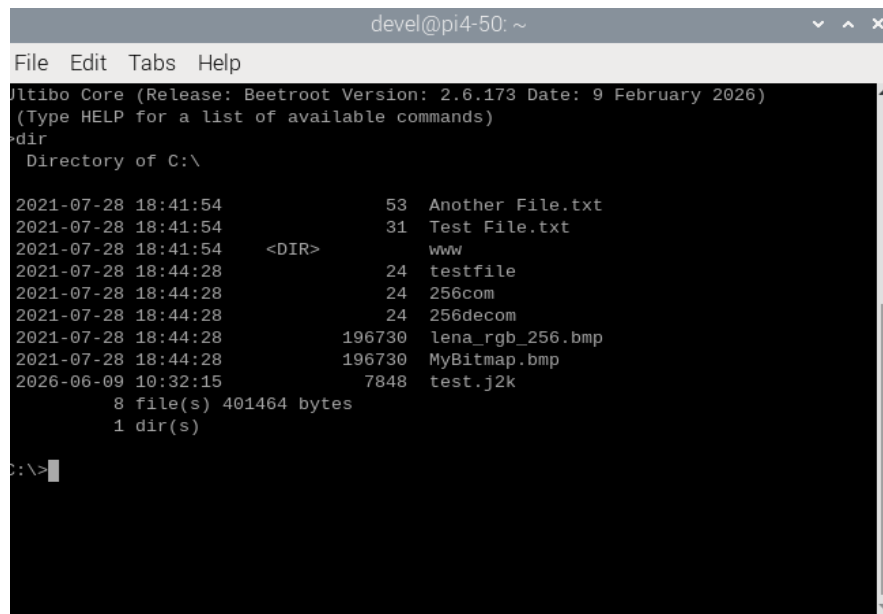
```
od -x 256com
00000000 0019 0000 0001 0000 0000 0000 0000 0000
00000020 0100 0000 0100 0000
CR 0019 0000
Perform compress 0001 0000
devel@pi4-50:~/Ultibo_Projects/jpeg2000/QEMU/tmp $ tftp pi4-50 5069
tftp> binary
tftp> get testfile
tftp> get 256com
tftp> get 256decom
tftp> quit
devel@pi4-50:~/Ultibo_Projects/jpeg2000/QEMU/tmp $ shasum *
8f06328e4016b235ab6c0acbcca000d2bccda1315 256com
98fb712d8639e310b3a4cec437fc41c141fddd13 256decom
8f06328e4016b235ab6c0acbcca000d2bccda1315 testfile
devel@pi4-50:~/Ultibo_Projects/jpeg2000/QEMU/tmp $ od -x 256decom
00000000 0019 0000 0000 0000 0000 0000 0000 0000
00000020 0100 0000 0100 0000
00000030
```

To create the test_wr.bmp 256decom needs to be copied to testfile.

```
devel@pi4-50:~ $ telnet pi4-50 5023
Ultibo Core (Release: Beetroot Version: 2.6.173 Date: 9 February 2026)
```

(Type HELP for a list of available commands)

>



available commands)

The QEMU, RPi, and RPi2 were rebuilt using the Lazarus IDE (ultibo-edition)

devel@pi4-50:~/Ultibo_Projects/jpeg2000/RPi2 \$ tftp 192.168.12.160

tftp> binary

tftp> put kernel7.img

tftp> quit

RPi2 reboots

devel@pi4-50:~/Ultibo_Projects/jpeg2000/src \$ cd ~/dwtlift-pi/pi-myTTDecoder/

devel@pi4-50:~/dwtlift-pi/pi-myTTDecoder \$ git diff

diff --git a/pi-myTTDecoder/myTTDecoder.c b/pi-myTTDecoder/myTTDecoder.c

index 9148dd6..a9c038d 100644

--- a/pi-myTTDecoder/myTTDecoder.c

+++ b/pi-myTTDecoder/myTTDecoder.c

@@ -95,8 +95,8 @@ printf("CR = %d flg = %d bp = %d imgsz = %d \n",CR, flg, bp, imgsz);

printf("da_x0 = %d da_y0 = %d da_x1 = %d da_y1 = %d\n",da_x0, da_y0, da_x1, da_y1);

@@ -106,4 +106,4 @@ printf("calling decom_test\n");

//decom_test(int da_x0, int da_y0, int da_x1, int da_y1, char *ff);

decom_test(da_x0,da_y0,da_x1,da_y1,ff);

return 0;

-}

\ No newline at end of file

*/

da_x0 = 0;

da_y0 = 0;

-da_x1 = 2048;

-da_y1= 2048;

```
+da_x1 = 256;  
+da_y1= 256;
```



xx